

HW8: Object Oriented Programming 2

Writeup: Texas Hold'em Poker

1. Game

Texas Hold'em Poker. Each player gets 2 hole cards, and 5 community cards get dealt out in stages (flop, turn, river). Players bet in rounds and can fold, check, call, raise, or go all in. Best 5-card hand out of the 7 available (2 hole + 5 community) wins the pot at showdown.

2. Objects in the Game

- Cards (rank + suit)
- A deck of cards
- A player's hole cards / hand
- Players
- The pot / chips
- The community cards ("the board")
- The dealer button / blinds

3. Classes

- `PlayingCard`: one playing card
- `CardDeck`: the 52-card deck, shuffling and dealing
- `PlayerHand`: a player's hole cards plus their best evaluated 5-card hand
- `BettingPot`: chips in the middle and the current bet to call
- `Participant`: a person at the table (chips, hand, betting actions)
- `HandRanker`: ranks a hand and breaks ties
- `TexasHoldEm`: runs the actual game loop (blinds, betting rounds, showdown)

4. Relationships

- `TexasHoldEm` has many `Participants`, one `CardDeck`, one `BettingPot`, and a list of `PlayingCards` for the board.
- Each `Participant` has one `PlayerHand`.
- Each `PlayerHand` has a list of `PlayingCards` (hole cards) and a list of `PlayingCards` (best 5-card hand once evaluated).
- `HandRanker` takes in a `PlayerHand` and the board and sets the hand's rank. It doesn't own any of them, just operates on them.

5. Attributes

Class	Attributes
PlayingCard	rank, suit
CardDeck	cards (list of PlayingCard)
PlayerHand	hole_cards, best_rank, best_five
BettingPot	chips, call_amount
Participant	name, chips, hand, wagered, folded, all_in
TexasHoldEm	participants, deck, pot, board, button, small_blind, big_blind, ranker

6. Methods

- PlayingCard: `__str__`, `art` (ascii-art card), `render_row` (prints a row of cards side by side)
- CardDeck: `shuffle`, `draw`
- PlayerHand: `deal_in`, `reset`, `record_result`
- BettingPot: `collect`, `clear`
- Participant: `wager`, `fold`, `award`, `start_new_hand`, `start_new_street`, `take_turn` (prompts for fold/call/check/raise/all-in)
- HandRanker: `best_of` (finds best 5-card hand and its rank), `score` (rates a 5-card hand), `has_straight` (checks for a straight, including the A-2-3-4-5 wheel), `compare` (breaks ties by kicker)
- TexasHoldEm: `run`, `play_one_hand`, `post_blinds`, `deal_hole_cards`, `run_betting`, `deal_board_cards`, `showdown`, `remove_busted`

Notes

- Code prints game state after every street (hole cards, community cards, chip counts, pot size) and at showdown.
- User input for moves is handled in `Participant.take_turn`.
- Dealer button and blinds rotate every hand; players with 0 chips get eliminated between hands.